

Movie Recommendation System Report

Name: Mansi Kapoor

College/Institute: Ideal Institute of Management And Technology

Internship Domain: AI And ML Interns

Project Duration: 45 Days

Introduction

The Movie Recommendation System is a machine learning project designed to suggest movies to users based on their interests and preferences. The system analyzes movie genres and user ratings to recommend similar movies. Recommendation systems are widely used in platforms such as Netflix, Amazon Prime, and YouTube to improve user experience and engagement. This project demonstrates how machine learning techniques can be applied to real-world applications using Python and Streamlit.

Abstract

This project uses the MovieLens dataset to build a movie recommendation application using content-based and collaborative filtering techniques. The system preprocesses movie and ratings data, converts movie genres into numerical vectors using TF-IDF vectorization, and calculates similarity scores using cosine similarity. The application then recommends the top five most similar movies to the user. A Streamlit-based user interface is also developed to provide an interactive experience.

Tools Used

Python: Used for programming and project development.

Pandas: Used for data loading, preprocessing, and manipulation.

Scikit-learn: Used for TF-IDF vectorization and cosine similarity calculations.

Streamlit: Used to create an interactive web interface for users.

MovieLens Dataset: Used as the dataset containing movie details and ratings.

Key Insights

The project showed that recommendation systems can effectively predict user interests by analyzing movie genres and rating patterns. Content-based filtering helps recommend movies with similar genres, while collaborative filtering improves recommendations using user behavior and ratings. The system successfully generated accurate movie suggestions and demonstrated the importance of machine learning in entertainment platforms.

Steps Involved in Building the Project

1. Downloaded and extracted the MovieLens dataset.
2. Loaded movies.csv and ratings.csv using Pandas.
3. Cleaned and preprocessed the dataset by removing missing values.
4. Converted movie genres into numerical vectors using TF-IDF vectorization.
5. Calculated cosine similarity between movies.
6. Built a recommendation function to return top 5 similar movies.
7. Developed a Streamlit user interface for movie selection and recommendations.
8. Tested the application and verified recommendation results.

Conclusion

The Movie Recommendation System successfully demonstrated the implementation of machine learning techniques for personalized recommendations. Using content-based and collaborative filtering methods, the system generated relevant movie suggestions efficiently. The project improved understanding of data preprocessing, similarity calculations, recommendation algorithms, and Streamlit application development. This project can be further enhanced by adding movie posters, sentiment analysis, and advanced recommendation algorithms.